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# Post-Middle Oligocene origin of paleomagnetic rotations in Upper Permian to Lower Jurassic rocks from northern and southern Peru

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**Abstract**

We report paleomagnetic data from 28 sites of Upper Permian to Lower Jurassic strata from northern and southern Peru. In northern Peru (6°S), a stable magnetic component from six Permo-Triassic sites passes fold and reversal tests. The overall mean pole agrees well with Late Permian to Triassic poles from cratonal South America, suggesting this part of Peru has experienced neither significant rotation nor latitudinal transport since the Permo-Triassic. In southern Peru (13 to 16°S), thermal demagnetization isolates stable magnetic components in 16 of 22 Upper Permian to Lower Jurassic sites collected along the transition between the Altiplano and the Eastern Cordillera. These 16 sites are rotated 14 to 147° counterclockwise and pass an inclinations-only fold test. Within the same structural zone, three other Permo-Triassic sites as well as 10 Paleocene sites also show important counterclockwise rotations [Roperch and Carlier, J. Geophys. Res. 97 (1992) 17233–17249; Butler et al., Geology 23 (1995) 799–802]. The large magnitude and exclusively counterclockwise sense of rotation suggest that the tectonic regime included an important sinistral shear component. No correlation exists between rotation amount and rock age, suggesting the rotations are post-Paleocene in age. Because the rotations occur along the fringe of the Eastern Cordillera, they were likely produced during its structural formation, hence from the Late Oligocene to Present. Sinistral shear acting in the northern part of the Bolivian Orocline appears much more pronounced than that north of the Abancay Deflection, which likely arises from differences in convergence obliquity.
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*Keywords:* Paleomagnetism; Andes; Altiplano; tectonics; Permian; Triassic; Jurassic; Peru; South America

## 1. Geologic setting and previous paleomagnetic work

The Andes, one of the world's longest and

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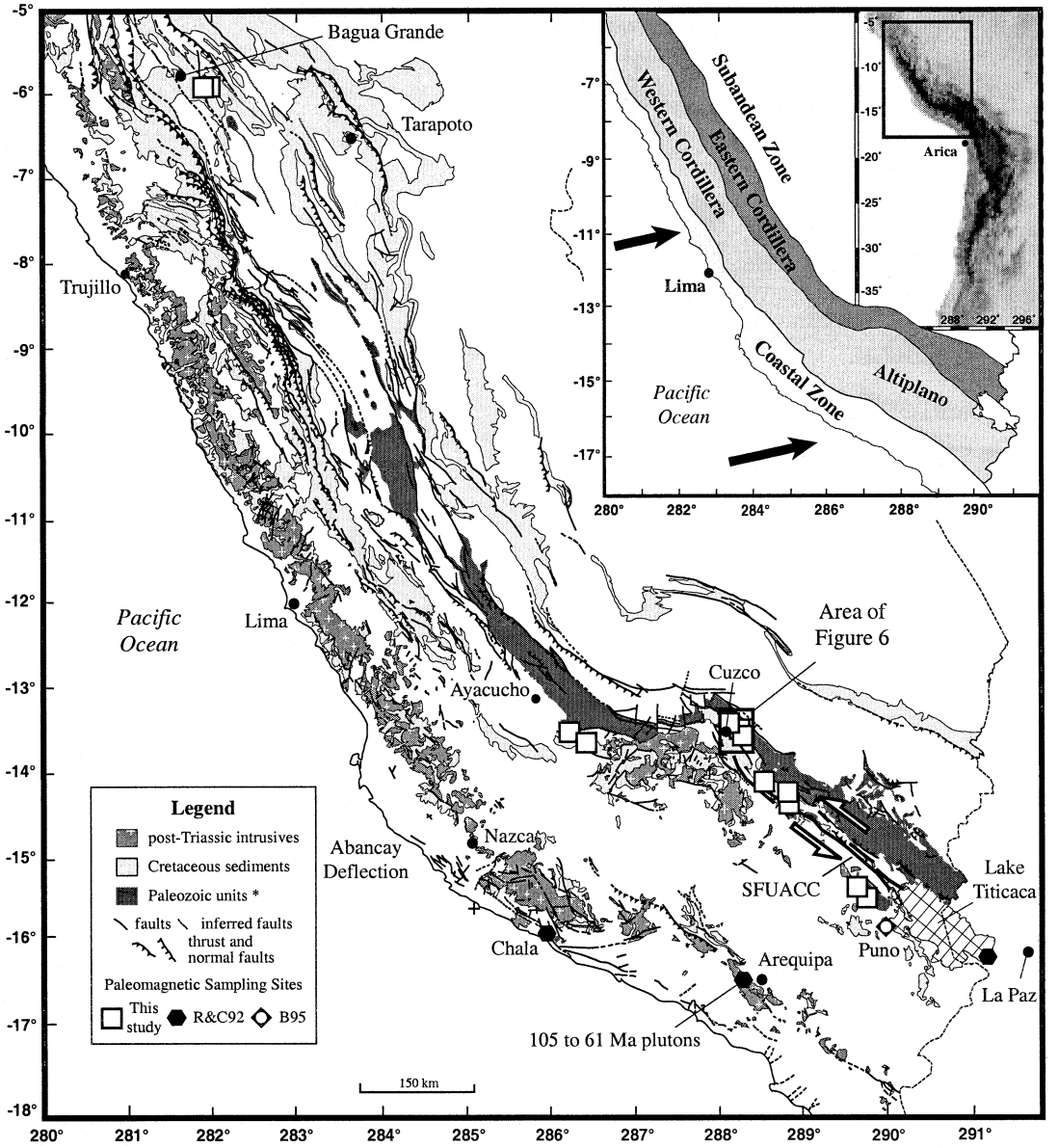



Fig. 1. Simplified geologic map of Peru showing sampling sites from this study and those from Roperch and Carlier [1] (R and C92) and from Butler et al. [2] (B95). Note that all localities from 13 to 16°S are counterclockwise rotated between 14 and 147°; those from the north (6°S) are not rotated. SFUACC is the left-lateral fault system running from Cuzco-Pisac through Lake Titicaca (see [18]); Paleozoic units are pre-Permian rocks that crop out in the Eastern Cordillera (simplified). Convergence directions (inset) calculated from [42]; convergence obliquities at 11 and 16.5°S latitudes are ca. 15 and 40°, respectively.

highest mountain ranges, span the entire length of the western margin of the South American plate (Fig. 1, inset). While their present morphology arises mainly from Late Cenozoic geodynamic processes [3], the geology recorded within the range infers a protracted history of deformation. The Peruvian segment of the Andes contains evidence of spatially variable Paleozoic and older

magmatism and metamorphism [4–7]. Where found, Upper Devonian to Lower Mississippian low-grade metamorphic rocks and intrusives [6,8] are unconformably overlain by Mississippian continental sediments and shallow marine Pennsylvanian to Lower Permian strata. Upper Permian to Lower Jurassic volcanics, red beds, and marine limestones, shales and sandstones accumulated in an extensional tectonic regime [6,7,9,10]. This Late Paleozoic to Early Mesozoic rifting event initiated in central Peru along what is today the Eastern Cordilleran axis (Fig. 1, inset) and propagated southwards into Bolivia. In the Puno region, Sempere et al. [10] observed that deformation of Jurassic (by folding) or Triassic (by cleavage) strata is post-dated by less deformed Upper Cretaceous deposits, thus documenting a latest Jurassic and/or Early Cretaceous deformation event there.

Subduction of oceanic crust under the South American plate possibly began as early as the Early Jurassic [11]. In the Arequipa Basin of southern Peru, deformation and related magmatism locally occurred in the latest Jurassic–earliest Cretaceous ([10] and references therein). The margin again underwent extension during most of the Early Cretaceous [12]. A significant evolution in tectonic mode occurred with the development of the Coastal Batholith, which marks the position of a Late Cretaceous–Paleogene magmatic arc [13] (Fig. 1). Compressional deformation is believed to have started in the westernmost part of the margin in the early Late Cretaceous [14], and marine shelf carbonates disappeared from the geologic record of Andean Peru around that time [12].

Unraveling the pre-Cretaceous tectonic evolution of the Peruvian Andes is obstructed by the lack of kinematic data. Existing paleomagnetic data (paleolatitudes) are coherent with those from the stable South American plate [1,15] and no geologic evidence exists to support the presence of exotic terranes in the Phanerozoic. In order to gain an understanding of the pre-Cretaceous tectonic processes affecting the region, we collected paleomagnetic cores mainly from the Early Mesozoic rift-related sediments exposed in the transition zone between the Altiplano and the Eastern Cordillera of the Peruvian Andes. We also sampled similar facies in northern Peru to compare with those from the south.

Pre-Cretaceous paleomagnetic data from Peru are limited to three studies. Creer [16] published two poles from the Mitu Group (considered at the time as Upper Carboniferous to Lower Permian) collected northeast of Lima. However, these data were not based on stepwise demagnetization and their reliability is questionable. Shackleton et al. [8] reported Jurassic data from four sites composed of andesitic lava flows and 17 sites from basaltic dikes. Unfortunately, the data can not be interpreted with confidence because the lock-in time of magnetic remanence was not established (no fold or reversals tests – all sites possess normal polarity only) and the mean direction of the four andesite sites has a very large uncertainty ($\alpha_{95} = 29°$).

Roperch and Carlier [1] published paleomagnetic results of 36 Jurassic through Cretaceous sites from the coast between 16 and 18°S and three Permo-Triassic sites near La Paz (Fig. 1). One post-folding magnetic component observed in most rocks was rotated $\sim 20°$ counterclockwise and was interpreted as a remagnetization, possibly due to a Late Cretaceous tectonomagmatic event. Several Jurassic to Cretaceous sites had primary magnetization components, which were associated with large counterclockwise rotations ($82.6 \pm 15.3°$ and $55.1 \pm 19.3°$ for Chala and Arequipa, respectively, Fig. 1). As the remagnetization component is rotated much less than the primary magnetic component, the authors concluded that the age of rotation of the primary component pre-dates the Late Cretaceous. The three sites of Permo-Triassic red beds from the La Paz region were also rotated counterclockwise, with a mean of $72 \pm 18°$. Magnetic components from other localities that may represent primary remanences were interpreted as preliminary and not considered further.

## 2. Paleomagnetic sampling and data treatment

We collected 258 paleomagnetic samples from 28 sites from the Upper Permian to Middle Tri-

Table 1
High-temperature component paleomagnetic data from this study

| Site | Age | $S_{lat}$ | $S_{long}$ | no/No | $N/R$ | $D_g$ | $I_g$ | $D_s$ | $I_s$ | $k$ | $\alpha_{95}$ | Stk | Dip | $P_{lat}$ | $P_{long}$ | dp | dm | Rot | ±Rot |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Northern Peru (Bagua Grande, 6°S)** | | | | | | | | | | | | | | | | | | | |
| P938 | P-T | ca. 200 m below P939 | | 8/8 | 0/8 | 172.2 | 8.5 | 168.7 | 42.9 | 65.2 | 6.9 | 272.1 | 35.0 | −68.1 | 310.4 | 5.3 | 8.5 | | |
| P939 | P-T | −5°55.13′ | 281°54.82′ | 8/9 | 8/0 | 353.7 | −4.2 | 348.6 | −39.0 | 38.4 | 9.1 | 280.8 | 36.7 | −70.5 | 315.1 | 6.5 | 10.9 | | |
| P940 | P-T | −5°55.49′ | 281°55.45′ | 8/8 | 8/0 | 346.6 | −7.2 | 350.9 | −31.9 | 24.5 | 11.4 | 233.3 | 27.3 | −75.5 | 319.1 | 7.2 | 12.8 | | |
| P941 | P-T | −5°55.16′ | 281°56.50′ | 6/8 | 6/0 | 340.9 | −33.4 | 349.7 | −48.1 | 223.6 | 4.5 | 220.4 | 17.8 | −64.8 | 303.4 | 3.8 | 5.9 | | |
| P942 | P-T | between P941 and P943 | | 8/8 | 0/8 | 186.3 | 45.5 | 184.2 | 42.4 | 101.9 | 5.5 | 60.1 | 3.8 | −70.9 | 270.1 | 4.2 | 6.8 | | |
| P943 | P-T | −5°54.86′ | 281°58.05′ | 7/8 | 0/7 | 10.6 | −9.6 | 353.7 | −36.1 | 141.6 | 5.1 | 328.4 | 45.5 | −74.6 | 304.7 | 3.4 | 5.9 | | |
| Mean | P-T | | | 6/6 | | 174.8 | 18.2 | – | – | 16.5 | 17.0 | | | | | | | | |
| Mean | P-T | | | 6/6 | | – | – | 172.6 | 40.2 | 125.5 | 6.0 | | | −71.4 | 303.6 | $A_{95} = 5.7$ | | −0.7 | 7.7 |
| **Southern Peru (13 to 16°S)** | | | | | | | | | | | | | | | | | | | |
| P832 | P-T | −15°26.27′ | 289°46.13′ | 8/8 | 0/8 | 72.4 | 39.4 | 80.8 | 53.6 | 74.7 | 6.5 | 133.0 | 18.0 | −1.3 | 344.6 | 9.1 | 6.3 | −93.2 | 8.9 |
| P833 | P-T | −15°21.13′ | 289°39.64′ | 8/8 | 0/8 | 108.7 | 11.1 | 98.5 | 50.5 | 81.4 | 6.2 | 216.4 | 42.3 | −15.0 | 350.8 | 8.3 | 5.6 | −75.5 | 8.5 |
| P836 | P-T | −15°18.60′ | 289°39.53′ | 8/8 | 0/8 | 80.6 | 38.7 | 115.3 | 36.3 | 58.5 | 7.3 | 95.0 | 42.5 | −28.6 | 64.7 | 8.5 | 5.0 | −58.7 | 8.8 |
| P838 | Jlq | −14°21.79′ | 288°50.45′ | 8/8 | 0/8 | 48.1 | −17.3 | 39.0 | 41.0 | 31.9 | 10.0 | 168.0 | 68.0 | 36.3 | 334.6 | 12.2 | 7.4 | −147.0 | 10.9 |
| P839 | Jlq | −14°15′* | 288°46′* | 7/8 | 7/0 | 4.9 | −22.9 | 352.5 | −32.8 | 19.4 | 14.1 | 336.0 | 25.0 | −82.0 | 351.9 | 16.0 | 9.0 | −13.5 | 13.6 |
| P840 | P-T | −14°06.48′ | 288°32.74′ | 11/11 | 11/0 | 241.3 | −25.4 | 226.4 | −33.9 | 93.8 | 4.7 | 216.0 | 27.0 | 33.8 | 344.3 | 5.4 | 3.1 | −127.5 | 7.4 |
| P841 | P-T | ca. 300 m from P840 | | 8/8 | 8/0 | 279.7 | −12.1 | 276.5 | −40.4 | 28.9 | 10.5 | 201.0 | 29.0 | −11.3 | 357.4 | 12.7 | 7.7 | −77.4 | 11.0 |
| P842 | P-J | −13°34.79′ | 288°17.18′ | 7/11 | 2/5 | 101.4 | 53.6 | 145.3 | 41.6 | 12.7 | 17.6 | 107.4 | 39.0 | −55.7 | 355.6 | 21.5 | 13.1 | −28.6 | 16.6 |
| P843 | P-J | ca. 30 m below P842 | | 6/10 | 6/0 | 271.2 | −41.1 | 301.6 | −40.1 | 19.6 | 15.5 | 104.7 | 34.0 | −34.1 | 359.7 | 18.7 | 11.3 | −52.3 | 14.8 |
| P844 | P-J | ca. 20 m below P843 | | 9/10 | 9/0 | 231.0 | −33.4 | 263.5 | −43.0 | 27.6 | 10.0 | 80.0 | 40.0 | 0.0 | 352.5 | 12.4 | 7.7 | −90.4 | 10.8 |
| P845 | P-J | −13°33.40′ | 288°16.97′ | 8/8 | 8/0 | 28.1 | −34.4 | 335.2 | −42.4 | 9.7 | 18.7 | 355.2 | 59.0 | −64.2 | 349.4 | 23.0 | 14.2 | −30.7 | 17.8 |
| P847 | P-J | −13°33.18′ | 288°16.73′ | 8/8 | 0/8 | 160.6 | 39.8 | 137.7 | 20.0 | 16.0 | 14.3 | 355.5 | 42.0 | −48.5 | 18.0 | 15.0 | 7.8 | −48.2 | 13.3 |
| P848 | P-J | −13°31.82′ | 288°15.53′ | 8/9 | 0/8 | 161.6 | 49.8 | 130.6 | 27.6 | 46.1 | 8.2 | 355.5 | 42.0 | −42.2 | 10.6 | 9.0 | 4.9 | −55.3 | 9.4 |
| P849 | P-J | −13°25.53′ | 288°08.32′ | 8/8 | 0/8 | 142.2 | 44.5 | 119.1 | 42.1 | 61.1 | 7.1 | 316.2 | 24.0 | −31.8 | 357.6 | 8.7 | 5.4 | −54.8 | 8.8 |
| P924 | P-T | −13°38.74′ | 286°25.59′ | 5/6 | 0/5 | 58.6 | −11.1 | 64.2 | 37.3 | 279.9 | 4.6 | 126.3 | 52.7 | 18.1 | 348.7 | 5.4 | 3.2 | −109.4 | 7.3 |
| P927 | Jlp | −13°31.76′ | 286°13.90′ | 7/11 | 0/7 | 75.1 | 26.2 | 113.7 | 41.3 | 36.7 | 10.1 | 108.5 | 54.0 | −26.9 | 356.2 | 12.3 | 7.5 | −72.0 | 11.0 |

Abbreviations are: $S_{lat}$, $P_{lat}$, $S_{long}$, $P_{long}$, latitude and longitude of the site (S) or pole (P); * estimated; no/No, number of samples used to calculate mean direction/number of samples demagnetized; $N/R$, number of samples with normal ($N$) and reversed ($R$) polarity; $D$, declination; $I$, inclination; g, in situ (geographic) coordinates; s, tilt-corrected (stratigraphic) coordinates (at 100% unfolding); $\alpha_{95}$,) the radius that the mean direction (pole site level mean) lies within 95% confidence; $k$, the best estimate of the precision parameter; Stk, strike (= dip direction minus 90°); dp and dm, semi-minor and semi-major axes of the 95% confidence oval on the pole calculated at the sample level; Rot, paleomagnetic rotation calculated using reference APWP (Table 2, 40 Ma window), negative values indicate counterclockwise rotation; ±Rot, 95% uncertainty on rotation value; P-T, Permo-Triassic Mitu Group; P-J, Mitu Formation to Lower Jurassic Quilcaunco (Caycay) Formation (Jlq); Jlp, Upper Triassic to Lower Jurassic Pucará Formation.

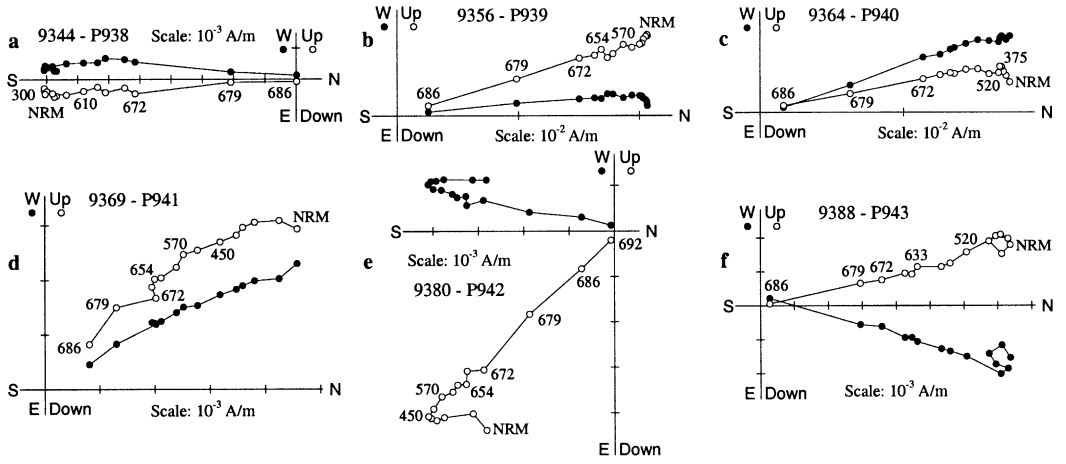



Fig. 2. Orthogonal demagnetization diagrams of Permo-Triassic rocks from the Bagua Grande region in in situ coordinates (sample number and corresponding sites provided). Demagnetization steps in °C; NRM is the natural remanent magnetization.

assic Mitu Group red beds and volcanics, the Upper Triassic to Lower Jurassic Pucará Group limestones, and the Upper Triassic to Lower Jurassic Quilcapunco (Caycay) Formation red sandstones (Table 1). In northern Peru, near the town of Bagua Grande (Fig. 1), we collected six sites of Mitu Group red beds, which are assigned an Upper Permian to Lower Triassic age by Sánchez [17]. Regionally, the fossil-poor Mitu Group conformably overlies the Lower Permian Copacabana Group and is unconformably overlain by the Late Triassic to Early Jurassic Pucará Group limestones, both of whose ages are paleontologically well constrained. In southern Peru, 22 sites come from a transect stretching from Puno to Cuzco (13 to 16°S) (Fig. 1). A description, including age assignments of these sediments was recently published by Sempere et al. ([10] and references therein). Most of the sites from southern Peru lie within the Urcos–Ayaviri–Copacabana–Coniri fault system (SFUACC in Spanish) (Fig. 1), which has been interpreted as a crustal to lithospheric left-lateral wrench system active in the Oligo-Miocene [18].

Paleomagnetic samples were collected with a gasoline-powered drill and oriented with magnetic and sun compasses, the latter to correct for local declination anomalies. Stepwise (averaging 18 steps) thermal demagnetization was applied to all samples. Magnetic remanences were measured with a 2G, three-axis DC SQUID magnetometer housed in a magnetically shielded room at the Institut de Physique du Globe de Paris. Magnetic components were determined using principal component analysis [19], and mean directions were calculated using Fisher [20] statistics. Data were treated using software developed by Cogné [21] and R. Enkin (unpublished).

Stepwise demagnetization of the Bagua Grande samples yielded a very stable magnetic component above 400 to 500°C that decays toward the origin, with about 50% of the remanence eliminated in the final 20°C before total unblocking by 690°C (Fig. 2; Table 1). The fold test statistic for this component maximizes at $104 \pm 19\%$ unfolding (Fig. 3) ([22] – using 1000 simulations and assuming a 5° uncertainty on bedding corrections), indicating a pre-folding age for the remanence. A sample level reversal test is positive at 95% confidence limits [23]. The magnetic stability and positive field tests suggest the high-temperature component represents a primary magnetization acquired in the Late Permian to Early Triassic.

Demagnetization of the samples collected from the Puno to Cuzco (13 to 16°S) transect isolated one to three magnetic components (Fig. 4). In samples exhibiting two components, declinations were northerly with shallow inclinations (in situ coordinates) in the low-temperature part of the unblocking spectra, probably representing a re-

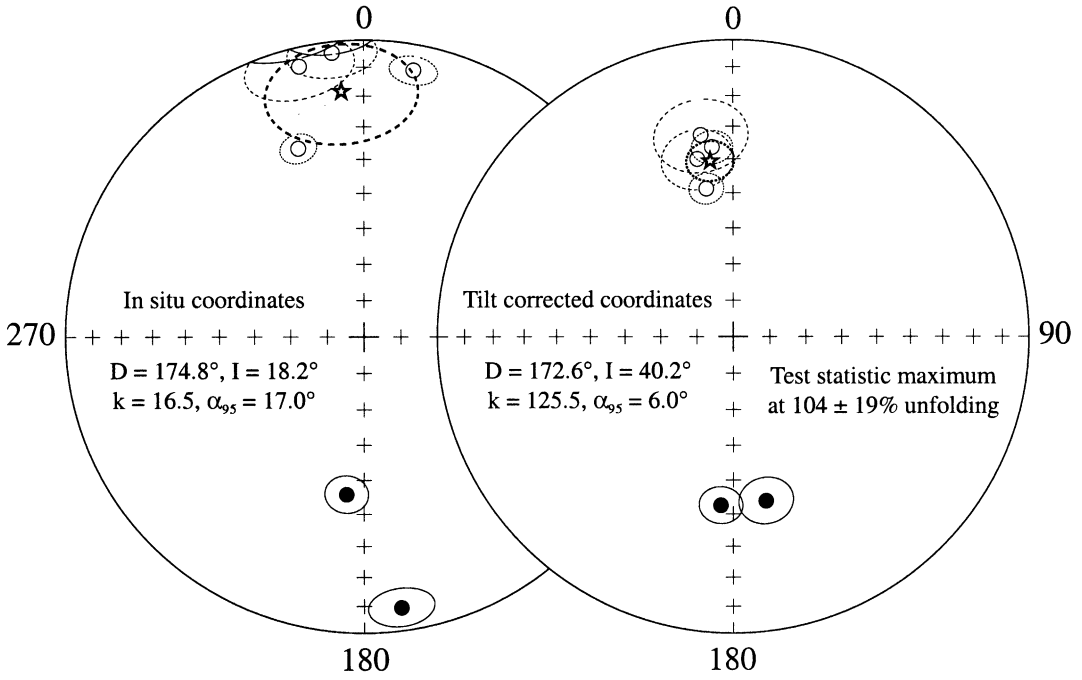



Fig. 3. Equal area stereonet plots of site mean directions and overall mean directions (stars) from the Permo-Triassic sites, Bagua Grande region (data from Table 1).

cent field direction. High-temperature components had easterly declinations when polarity was reversed, and westerly declinations when polarity was normal; inclinations were generally much steeper than those of the low-temperature component. Based on unblocking temperatures, the remanence was carried predominantly by hematite in the red beds (although sometimes magnetite, see Fig. 4d) and magnetite in the limestones (Fig. 4n,o). Sample directions with three magnetic components were generally highly dispersed, often resulting in site mean directions with $\alpha_{95} > 20°$. These sites (six of 22) were omitted from further consideration. For the other 16 sites, mean directions were calculated from the high-temperature magnetic components (Fig. 5; Table 1). Declinations of these sites were quite variable. However, the inclinations were much better grouped in tilt-corrected than in in situ coordinates (mean $I_{\text{geographic}} = 31.3 \pm 9.0°$, $k = 17.4$; mean $I_{\text{stratigraphic}} = 39.9 \pm 5.4°$, $k = 44.4$), suggesting that the magnetic remanences were acquired before folding (note that Late Permian to Middle Jurassic expected inclinations from the South American apparent polar wander path are similar, see below). Of all the sites from southern Peru, P839, collected on the road going east from Sicuani to the village of Uyurmiri, is the least rotated. As all the in situ sample directions from this site are of normal polarity and are not significantly different from the geocentric axial dipolar field direction ($D = 0°$, $I = -26.9°$), they could represent a remagnetization in the Brunhes field; remagnetization in the present field direction ($D = -2.6°$, $I = -3.0°$) can be ruled out (Fig. 5).

To better understand the origin of the dispersion in declinations, we first examined the data from the Pisac-Cuzco region (Fig. 6). Of the seven sites, six possess relatively coherent tilt-corrected directions, whereas one (P844) shows a more westerly declination than the rest. This latter site is comprised of basalt, whereas the others are red beds. The red bed sites P842 and P843 overlie P844 by 20 and 50 m, respectively, and possess

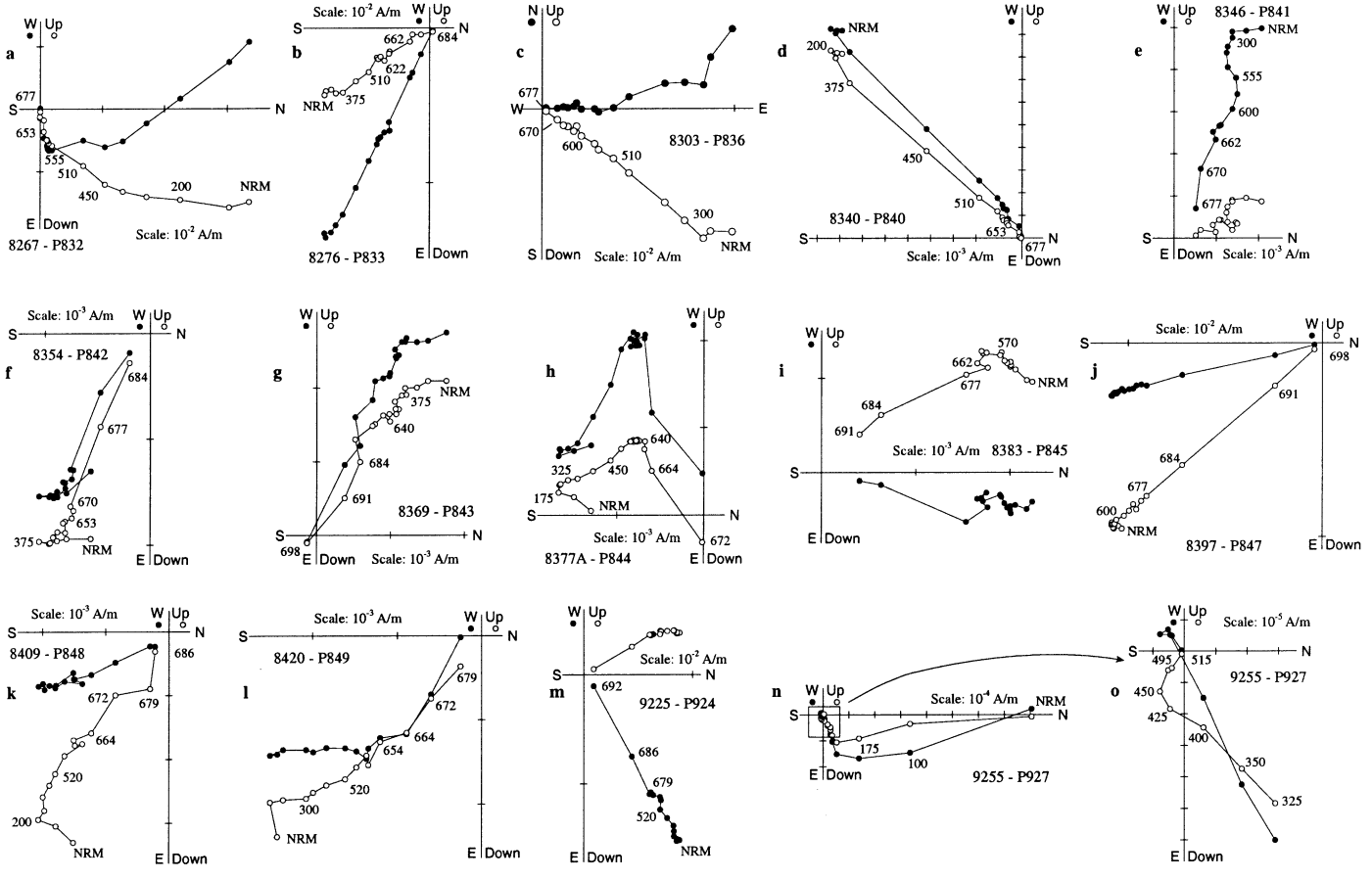



Fig. 4. Orthogonal demagnetization diagrams of rocks from southern Peru in in situ coordinates (sample number and corresponding sites provided). Demagnetization steps in °C; NRM is the natural remanent magnetization.

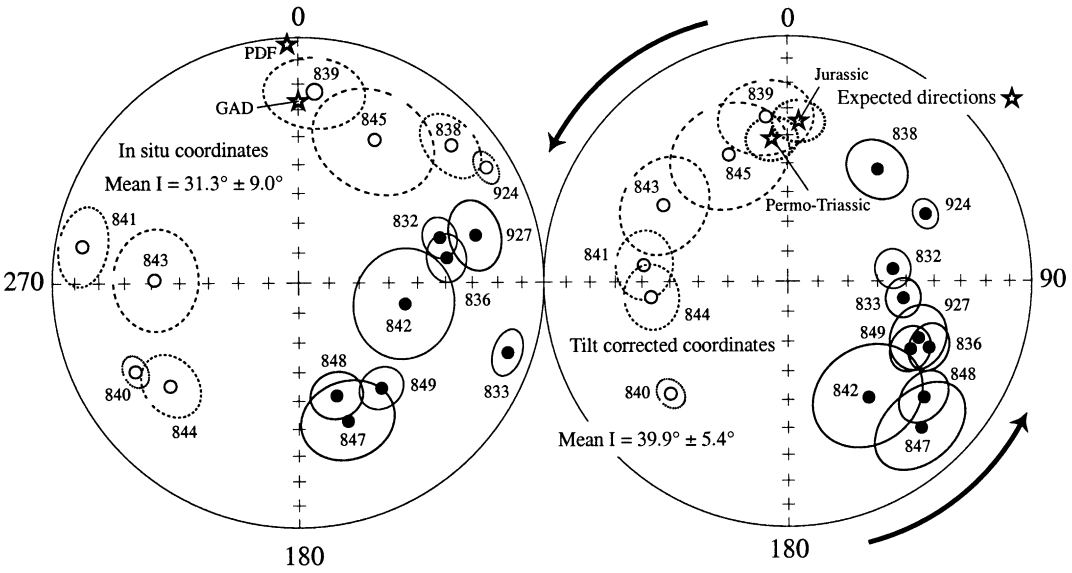



Fig. 5. Equal area stereonet plots of site mean directions from sites located in southern Peru (data from Table 1). Stars are expected directions calculated from the Late Permian to Early Triassic (247 ± 23 Ma) and Late Triassic to Early Jurassic (190 ± 26 Ma) poles from the 40 Myr window in Table 2; GAD is the geocentric axial dipole field direction and PDF is the present day field direction calculated for a point at −14°S, 289°E.

slightly different bedding attitudes (Table 1). Because we could not determine whether the basalt was a volcanic flow or a dike, we assigned its strike and dip by measuring the plane normal to columnar joints within the basalt. If one assigns the bedding orientations from the overlying red bed sites to P844, its tilt-corrected site mean direction remains significantly different from the others; thus, the more westerly declination is not an artefact of an erroneous bedding correction. As the thickness of the basalt is about 10 m, the most likely explanation for the deviated direction of site P844 is that geomagnetic secular variation was not sufficiently averaged. Putting this site aside, the fold test statistic of the six other site mean directions maximizes at 99 ± 23% unfolding, which together with the presence of dual polarities indicates a pre-folding age for the magnetic remanence.

Not far from Puno, sites P832, P833 and P836 were collected in close proximity and have fairly homogeneous directions, which are also similar to those located east of Lake Titicaca (Fig. 1) [1]. Farther north, sites P840 and P841 were collected about 300 m apart, yet their declinations differ by 50°. On the other hand, site P838, located near P840 and P841, has a tilt-corrected direction very similar to P840, whereas the tilt-corrected direction of P841 resembles those farther south (P832, P833 and P836). Given the high density of faults in the region, including faults that are mapped cross-cutting the dominant regional trend of ca. 320°N (e.g., Fig. 6), it is not surprising that declinations are widely scattered due to differential vertical axis block rotations. With few exceptions, the coherence in declination over limited distances together with dual polarity magnetic remanence and a positive inclination test [24] on all the data suggest that the high-temperature components in Table 1 represent primary magnetic remanences.

## 3. Discussion and conclusions

In order to interpret our results we first constructed a Permian to Middle Jurassic apparent polar wander path (APWP) for the South American plate. This proved to be non-trivial because some of the data do not meet modern standards

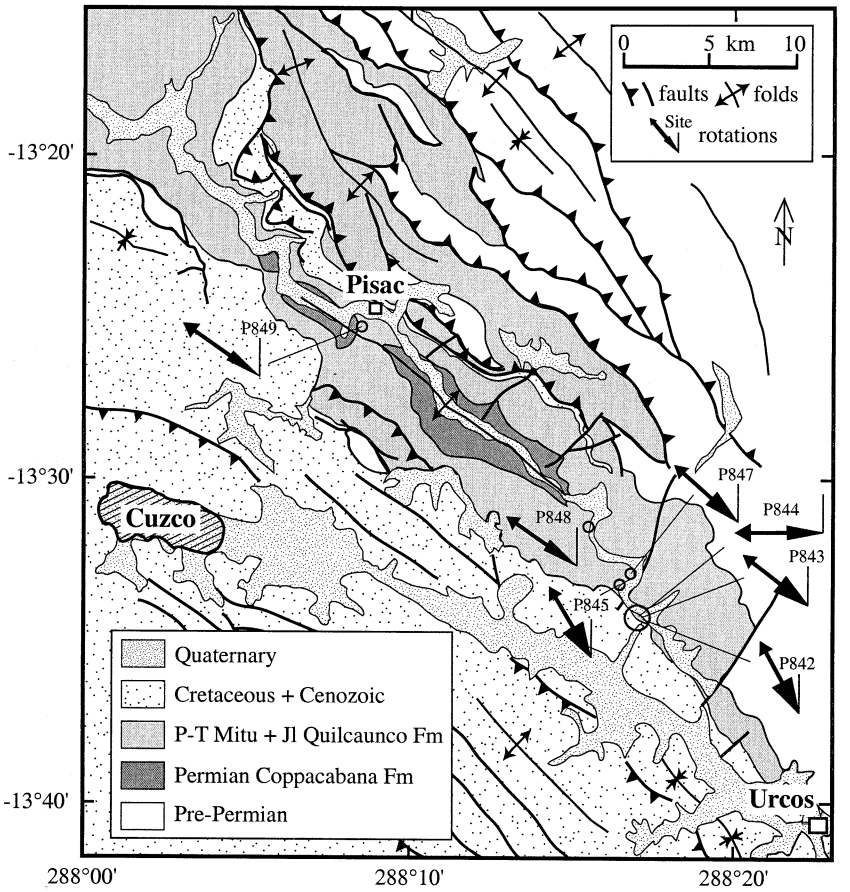



Fig. 6. Simplified geologic map of the Cuzco-Pisac–Urcos region (after [43,44]). Note that site P844 is composed of basalt while the other sites are red beds. P-T is Permo-Triassic and Jl is Lower Jurassic.

of reliability (no demagnetization treatment, pole $A_{95} > 15°$, etc.), whereas in other cases, poles are reported with no details surrounding demagnetization treatment, field tests, etc. We omitted from our compilation those data (1) obtained west of the Eastern Cordillera of the Andes around the Bolivian Orocline ($\sim$19°S) because of the plethora of rotations found in Cretaceous to Present rocks there (e.g., [25]), (2) from studies where the rocks were concluded to have undergone vertical axis block rotation (e.g., [26]), or (3) derived from plutons (where paleohorizontal is unknown) (e.g., [27]). We accepted those results when we could recalculate the tilt-corrected directions and poles from the in situ directions and bedding attitudes, verify the mean values, and when possible, determine the percentage of unfolding where the test statistic or precision parameter was maximized.

The South American plate reference poles we retained/recalculated are listed in Table 2 and shown in Fig. 7 (note southern hemisphere projections) with their corresponding localities plotted on the map to the right. In general, there is a good spatial representation for each time period with a tendency for pole position to change systematically with time. From these poles, we calculated two APWPs by fitting sliding windows with widths of 30 and 40 million years (Table 2). A trade-off exists in that smaller windows have better time resolution but larger uncertainties ($A_{95}$) on pole positions, whereas larger windows have poorer time resolution but smaller $A_{95}$. The im-

Table 2
South American plate reference paleomagnetic poles

| # | Age | Age (Ma) | $S_{lat}$ | $S_{long}$ | $P_{lat}$ | $P_{long}$ | $A_{95}$ | $N$ | Reference | Comment |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Jm | 146–170 | −6.5 | 318.0 | −83.6 | 81.0 | 1.9 | 21s | Schult and Guerreiro [45] | a |
| 2 | Jm | 161 | −48.2 | 291.2 | −84.8 | 173.8 | 7.9 | 37s | Vilas [46] | b |
| 3 | Jm | 161–171 | −44.9 | 294.3 | −82.0 | 309.0 | 12.0 | 17s | Creer et al. [47] | c |
| 4 | Jm | 168–173 | −41.8 | 289.8 | −70.2 | 190.4 | 9.7 | 13s | Rapalini and Lopez de Luchi [48] | d |
| 5 | Jl | 190–196 | −43.0 | 289.3 | −75.6 | 129.4 | 6.7 | 13s | Vizan [49] | e |
| 6 | Jl | 192–198 | 4.2 | 307.2 | −81.2 | 235.1 | 4.0 | 26s | Nomade et al. [50] | f |
| 7 | Jl | 197 | −13.6 | 300.8 | −65.5 | 250.8 | 3.9 | 14s | Montes-Lauar et al. [51] | g |
| 8 | Tu-Jl | 195–204 | 8.3 | 294.3 | −70.7 | 271.3 | 8.3 | 10s | McDonald and Opdyke [52] | h |
| 9 | Tu | 227 | 4.0 | 305.0 | −81.5 | 320.3 | 5.8 | 12s | Veldkamp et al. [53] | i |
| 10 | Tm-Tu | 219–236 | −32.7 | 292.2 | −77.9 | 237.8 | 10.4 | 15s | Valencio et al. [54] | j |
| 11 | Pu-Tl | 242–256 | −5.9 | 281.9 | −71.4 | 303.6 | 5.7 | 6s | this study, Bagua Grande | k |
| 12 | P-K | 260–290 | −32.0 | 290.5 | −79.4 | 308.7 | 6.8 | 16s | Rapalini and Vilas [55] | l |
| 13 | P-K | 260–290 | −38.0 | 298.2 | −65.9 | 20.7 | 7.0 | 24s | Tomezzoli and Vilas [56] | m |
| 14 | P-K | 260–290 | −20.8 | 296.2 | −81.8 | 344.2 | 3.5 | 60s | Libarkin et al. [32] | n |
| 15 | P-K | 260–290 | −41.5 | 294.7 | −74.7 | 310.5 | 7.3 | 13s | Rapalini [57] | o |
| 16 | P-K | 260–290 | −38.2 | 298.6 | −74.1 | 25.9 | 8.0 | 24s | Tomezzoli [58] | p |
| 17 | Pl | 266–290 | −30.0 | 293.0 | −84.0 | 12.4 | 9.2 | 5S | mean of five studies | q |
| South American Plate Reference Poles | | | | | | | | | | |
| | Jm | 164 ± 6 | | | −85.3 | 188.8 | 13.8 | 4S | 30 Myr sliding window | |
| | Jl-Jm | 187 ± 15 | | | −79.8 | 233.1 | 12.4 | 6S | 30 Myr sliding window | |
| | Tm-Jl | 207 ± 16 | | | −80.2 | 248.1 | 11.2 | 6S | 30 Myr sliding window | |
| | Pu-Tu | 235 ± 14 | | | −79.0 | 288.4 | 15.5 | 3S | 30 Myr sliding window | |
| | Pl | 276 ± 1 | | | −78.7 | 355.2 | 8.5 | 6S | 30 Myr sliding window | |
| | Jl-Jm | 177 ± 17 | | | −83.3 | 208.1 | 10.2 | 7S | 40 Myr sliding window | |
| | Tu-Jm | 189 ± 25 | | | −83.0 | 234.7 | 7.8 | 10S | 40 Myr sliding window | |
| | Pl-Tm | 256 ± 28 | | | −80.3 | 324.3 | 7.2 | 10S | 40 Myr sliding window | |
| | Pl | 272 ± 10 | | | −78.5 | 344.8 | 8.1 | 7S | 40 Myr sliding window | |

Abbreviations are: #, pole number in Fig. 7; J, Jurassic; T, Triassic; P, Permian; l, Lower; m, Middle; u, Upper; P-K, in Permian part of Kiaman reversed polarity superchron; $S_{lat}$ ($P_{lat}$), mean site (pole) latitude; $S_{long}$ ($P_{long}$), mean site (pole) longitude; $A_{95}$, radius of 95% confidence about the mean pole; $N$, number of sites (s) or studies (S) used in the mean pole position. Comments are: (a) no structural correction, only one demagnetization step applied in most cases; (b) only one of 34 sites had reversed polarity, supersedes Valencio and Vilas [59]; (c) only one demagnetization step applied in most cases, no structural correction; (d) dykes only, no structural correction; (e) fold test positive, precision parameter maximum ($k_{max}$) at 100% unfolding; (f) combined A and B populations; (g) all directions are of normal polarity only, less than 40% of samples were included in study; (h) only one step demagnetization applied; (i) no structural correction; (j) only one demagnetization step applied to most samples, mean of 224 ± 5 Ma igneous rocks from the Los Rastros and Ischigualasto formations and 232 ± 4 Ma igneous rocks from the Puesto Viejo Formation; (k) see text; (l) study of Permian Tambillos Formation lies within Choiyoi Group, $k_{max}$ at 100% unfolding; (m) three combined sections yield syn-folding (40% unfolding) magnetization. Because folding affects Permian strata and all samples have reversed polarity, age of magnetization should be in Permian part of Kiaman; (n) re-interpretation of data in following way: the Devonian Vilavila Formation, sediments from Pongo fail the fold test statistic maximum ($ts_{max}$) at −15 ± 5% unfolding. Fold test performed on rest of sites shows $ts_{max}$ at 99 ± 3% unfolding. Because most sites have reverse polarity, magnetization was likely acquired in Kiaman part of Permian; (o) $ts_{max}$ at 16 ± 3% unfolding suggests syn-folding remagnetization, because all directions have reverse polarity, magnetization likely acquired in Permian part of Kiaman; (p) Early Permian red beds collected in same place as in Tomezzoli and Vilas [56]. The main difference between the two studies is that Tomezzoli and Vilas [56] report data from the Tunas I Formation, whereas those from Tomezzoli [58] come from the younger Tunas II Formation. Tunas II has $k_{max}$ at 90% unfolding whereas Tunas I has $k_{max}$ at 40% unfolding. All rocks contain reverse polarity; (q) several early studies from same formations at same localities. Here we take mean of five studies from Valencio et al. [60] (supersedes Sinito et al. [61]) and Thompson [62] ($k_{max}$ at 100% unfolding, supersedes Embleton [63]). Note that ages were reported in [64].

portant point for this study is that expected paleolatitudes and declinations for sites located in Peru change little (±10°) from the Late Permian through the Middle Jurassic (Fig. 7, bottom left), and thus our interpretations are relatively insensitive to both the age uncertainties in the formations we sampled and to which reference pole is used. Finally, our new poles agree very well with the South American reference poles proposed by Beck et al. [28], Randall [29], Van Der Voo [30] (except Lower Permian, see below) and Besse and Courtillot [31] (Fig. 7, bottom right).

One aspect of our compilation is that pole #17 (Table 2) represents a Fisherian mean of five individual studies from the region around 30°S, 293°E as to minimize the influence of data obtained from the same rocks collected in the same place, thus giving it the same weight as individual poles from other areas. Another important aspect is the addition of data from Libarkin et al. [32], who concluded that 74 sites of Devonian to Permo-Triassic rocks were remagnetized during Cenozoic folding. Our re-analysis of their data shows that the test statistic maximum ($ts_{max}$) [22] of all 74 sites occurs at 93±3% unfolding, consistent with a syn-folding magnetization. Further investigation finds that $ts_{max}$ for the Vilavila Formation from the Pongo locality occurs at −15±5% unfolding, suggesting the remanence is post-folding. The rest of the sites ($N=57$) (excluding three sites with $\alpha_{95}>40°$) have $ts_{max}$ at 99±3% unfolding, demonstrating a pre-folding remanence. As 52 of the 57 sites have reverse polarity, it is probable that the remanence was acquired mostly during the Kiaman (Late Carboniferous to early Late Permian) reversed polarity superchron, consistent with the age of the rocks. Roperch et al. [33] also concluded the predominantly reversed polarity was consistent with acquisition of magnetic remanence during the Kiaman and not during the Tertiary when the reversal frequency was high. They suggested the Libarkin data underwent a significant (~30°) clockwise rotation in the Cenozoic. However, the interpretation of these data is sensitive to which reference pole is used. When compared against the 274±7 Ma pole from Van Der Voo [30], which is displaced much farther away from the Kiaman pole calculated here via a sliding window (Fig. 7, bottom right), it is indeed rotated ~30°. On the other hand, one sees that the Libarkin pole (#14) lies well within the population of the individual Kiaman-aged poles retained in our compilation (Fig. 7), which argues against any significant rotation.

Fig. 7 shows the overall mean tilt-corrected Bagua Grande Mitu Group pole (pole #11) along with the other South American poles. The Mitu Group pole lies within the cluster of coeval poles, suggesting that the Bagua Grande area has experienced neither important rotation nor translation since the Permo-Triassic. It is worthwhile to note that about 200 km to the north of Bagua Grande, paleomagnetic vectors in Cretaceous to Paleogene rocks are rotated consistently clockwise [34] whereas about 200 km to its south, the vectors are rotated counterclockwise [34,35]. This implies that Bagua Grande is situated within a transition region between different tectonic regimes responsible for the rotations. Our previous work in the large area south of Bagua Grande (from 7 to 15°S) documents ~15° counterclockwise rotations in Cretaceous to Upper Cenozoic rocks [36,37]. Thus, the Bagua Grande area marks the northern limit of this large affected area. Bagua Grande also marks the region where the topography of the Andes becomes more subdued (less than 3000 m).

For each of our sites from southern Peru, we calculated the rotation and paleolatitude differences using the 256±28 and 189±25 Ma poles from the 40 Myr window APWP (Table 1; [38]). Neither the Permo-Triassic nor the Jurassic sites exhibit significant poleward displacement on average (2±11° and 1±10°, respectively). However, apparent in both Fig. 5 and Table 1 is that each site collected for this study from southern Peru is rotated counterclockwise with respect to stable South America. This is consistent with the three Permo-Triassic sites reported from the La Paz area ([1]; Fig. 1). Near Puno (Fig. 1), Butler et al. [2] studied the paleomagnetism of the Paleocene Laguna Umayo Formation and found this series to be rotated 33±10° counterclockwise. Although the overall mean direction is less rotated than the Permian to Jurassic strata from

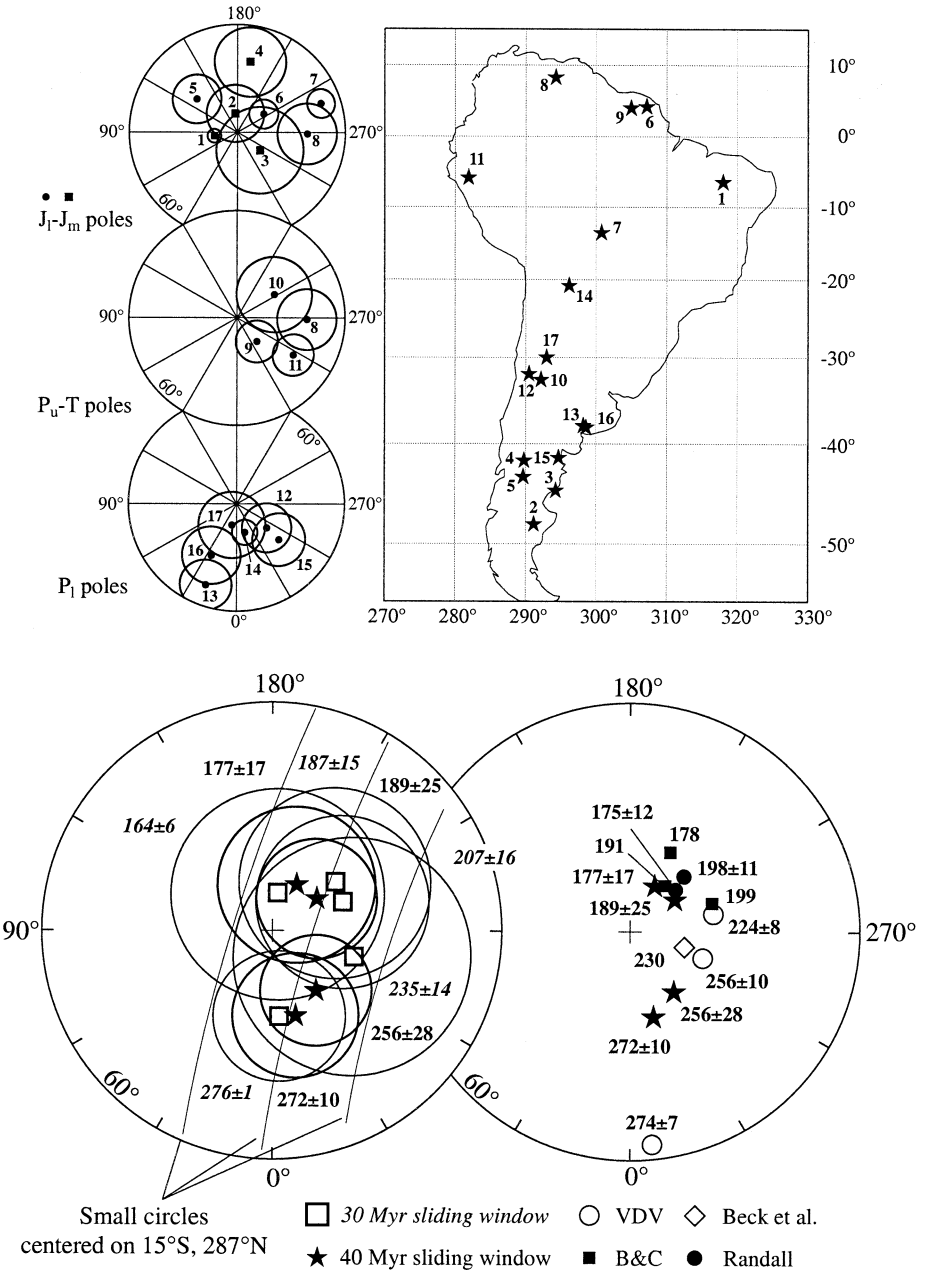



Fig. 7. Southern hemisphere projections of paleomagnetic poles from stable South America (numbered study localities on map and on pole projection correspond to data in Table 2). Plot in lower left shows the reference South American plate poles using 30 and 40 million year sliding windows; small circles plotted at 60, 70 and 80° from center. Plot in lower right shows the reference South American poles proposed by Randall [29], Beck [28], Van Der Voo (VDV) [30] and Besse and Courtillot (B&C) [31], together with poles from the 40 million year sliding window. Abbreviations: P, Permian; T, Triassic; J, Jurassic; u, Upper; m, Middle; l, Lower.

the same region, the individual site mean directions show large variations in declination (see fig. 2 in [2]), with corresponding rotations up to 80° counterclockwise, similar to the rotations of the older rocks (Fig. 1). Thus, because the rotations in Permo-Triassic to Paleocene rocks are similar in sense and magnitude, the age for these rotations is most likely post-Paleocene. Furthermore, although some deformation has occurred in this region as early as the latest Jurassic and/or earliest Cretaceous [10], the major deformation event started in the Late Oligocene [39], which is believed responsible for the formation of the Eastern Cordillera by tectonic inversion and shortening of this previously weakened area [18]. This would mean that the mechanism causing the Paleocene rotations observed by Butler et al. [2], and hence the rotations recorded in the Permian to Jurassic rocks from the same area, took place largely between the Late Oligocene to Present.

A remaining question is whether the deformation responsible for these rotations was restricted to the transition area between the Altiplano and Eastern Cordillera, or did it affect a larger area, such as in the Western Cordillera or Coastal Zone (Fig. 1, inset)? Roperch and Carlier [1] found that seven Middle Jurassic sites from the Chala region of southern coastal Peru were rotated 79 ± 15° counterclockwise (rotation recalculated based on seven sites, excluding site from pluton; Fig. 1), consistent with the rotations in the Eastern Cordillera. Moreover, near Arequipa (Fig. 1), these workers sampled four sites in plutons dated from 105 to 61 Ma that exhibit large (55 ± 19°) counterclockwise rotations, potentially placing the age of rotation younger than ~61 Ma. On the other hand, one can not exclude the possibility that the Arequipa gabbros were tilted, thus yielding declinations that are coincidentally rotated counterclockwise (as no tilt correction was applied), or that they are actually Middle Jurassic, consistent with the ages of the more mafic plutons found in the area [40]. In the same study, Roperch and Carlier [1] identified a widespread remagnetization component and proposed that it was imprinted in the Late Cretaceous. As the remagnetization direction is counterclockwise rotated by about 20°, the much larger counterclockwise rotations in the Middle Jurassic rocks should be older than the Late Cretaceous.

Emerging results from the coast (~17°S), however, show that the Lower Moquegua (Oligocene) Formation is rotated up to 45° counterclockwise, suggesting that the remagnetization component post-dates the Oligocene [41]. An even younger age for the remagnetization is thus called for, which is probably Upper Neogene, as all Upper Cretaceous to Upper Neogene rocks north of the Abancay Deflection (Fig. 1) from 15 to 7°S are consistently rotated about 15° counterclockwise [36,37]. The new results from the Lower Moquegua Formation leave the question open whether (1) a certain amount of counterclockwise rotation was acquired between the Early Cretaceous and the Oligocene in the southern coastal domain, or (2) further sampling will show even greater rotation amounts, thus placing all rotation in the time window of post-Oligocene to pre-Upper Neogene (i.e., Miocene). The presence of large counterclockwise rotations in plutons dated at 61 Ma and in Late Oligocene sediments suggests a similar timing with the rotations in the Eastern Cordillera. Taken together, the paleomagnetic data provide evidence that the Peruvian margin experienced little or no deformation resulting in vertical axis rotation throughout the Mesozoic and much of the Cenozoic. The youth of the rotations would suggest that they are due to the tectonic regime responsible for the development of the modern Andes, including the Bolivian Orocline.

Our sites from southern Peru lie in the transition area between the Altiplano and Eastern Cordillera (Fig. 1), which forms part of a major left-lateral wrench system (SFUACC, Fig. 1) [18]. The paleomagnetic results are compatible with the presence of left-lateral shear in that not only do we find systematic counterclockwise rotations, but that rotation magnitude varies along strike, which is commonly observed in strike-slip systems. Thus, while shortening has undoubtedly been a major factor in constructing the modern Andes, transpression has clearly played an important role in the northern Altiplano region. The large affected area calls for large-scale tectonic forcing. On the other hand, the agreement in paleolatitudes of these sites with those from cratonal

South America suggests this region is not allochthonous, consistent with the conclusions of Palmer et al. [15] and Beck et al. [27] for regions farther south. Because paleomagnetic rotations south of the Abancay Deflection are generally much greater ($>25°$) than those to the north ($<25°$), and because the rotations are a relatively young phenomena (Late Oligocene to Present), their source probably reflects differences in convergence obliquity, today being much higher ($\sim 40°$) south of the deflection than to its north ($\sim 15°$) (Fig. 1, inset) [42]. It is possible that this sinistral shear exploited the major lithospheric heterogeneity created by rifting in the Triassic that pre-conditioned the Eastern Cordillera region to facilitate subsequent deformation. If true, then the dominant control on the building of this mountain chain owes its origin to an older tectonic feature.

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